

Pre-damage biomass allocation and not invasiveness predicts tolerance to damage in seedlings of woody species in Hawaii

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Abstract. Plant-herbivore interactions have been predicted to play a fundamental role in plant invasions, although support for this assertion from previous research is mixed. While plants may escape from specialist herbivores in their introduced ranges, herbivory from generalists is common. Tolerance traits may allow non-native plants to mitigate the negative consequences of generalist herbivory that they cannot avoid in their introduced range. Here we address whether tolerance to herbivory, quantified as survival and compensatory growth, is associated with plant invasion success in Hawaii and investigate traits that may enhance tolerance in seedlings, the life stage most susceptible to herbivory. In a greenhouse experiment, we measured seedling tolerance to simulated herbivory through mechanical damage (50% leaf removal) of 16 non-native woody plant species differing in invasion status (invasive vs. non-invasive). Seedlings were grown for 2 weeks following damage and analyzed for biomass to determine whether damaged plants could fully compensate for the lost leaf tissue. Over 99% of all seedlings survived defoliation. Although species varied significantly in their levels of compensation, there was no consistent difference between invasive and non-invasive species. Seedlings of 11 species undercompensated and remained substantially smaller than control seedlings 2 weeks after damage; four species were close to compensating, while one species overcompensated. Across species, compensation was positively associated with an increased investment in potential storage reserves, specifically cotyledons and roots, suggesting that these organs provide resources that help seedlings re-grow following damage. Our results add to a growing consensus that pre-damage growth patterns determine tolerance to damage, even in young seedlings which have relatively low biomass. The lack of higher tolerance in highly invasive species may suggest that invaders overcome herbivory barriers to invasion in other ways, such as resistance traits, or that herbivory does not play an important role in the seedling invasion dynamics of these woody species in Hawaii.

Key words: *biomass allocation; compensation; Hawaii; invasive species; overcompensation 30 years later; seedling; tolerance.*

INTRODUCTION

Biological invasions are widely recognized as one of the major threats to biodiversity and ecosystem stability at local, regional, and global scales (Vitousek et al. 1997, Lockwood et al. 2007, Ehrenfeld 2010). Plant invasions in particular can impact community structure (Pyšek and Pyšek 1995, Dunbar and Facelli 1999), fire regimes (Hughes et al. 1991, Brooks et al. 2004), hydrology (DiTomaso 1998, Dyer and Rice 1999), and nutrient cycling (Vitousek and Walker 1989, Evans et al. 2001). Yet, even though non-native plants are transported

around the globe at an astonishing rate, only a small percentage of these introduced plants establish sustainable populations and become invasive (Richardson et al. 2000), and it remains unclear why some species become invasive while others do not. One reason for failure is biotic resistance, provided by native natural enemies that attack non-native plants in their introduced ranges (Levine et al. 2004, Bufford et al. 2016). Herbivory, as an important component of biotic resistance, has received considerable attention in the invasion literature (Chun et al. 2010). Herbivory mediates plant invasions in two ways. First, herbivory may prevent potentially invasive species from establishing by reducing plant growth and survival to such low levels that populations never establish. This type of biotic resistance leading to failed invasion before naturalization is often difficult to detect (Zenni and Nuñez 2013). Second, herbivory may act to regulate established populations by reducing non-native plant fitness, thereby limiting population growth

Manuscript received 28 February 2017; revised 10 August 2017; accepted 11 September 2017. Corresponding Editor: Ken N. Paige.

Editors' Note: Papers in Special Features are published individually and will be linked online in a virtual table of contents at: www.wiley.com/go/ecologyjournals

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and spread (Levine et al. 2004, Schmitz 2008, Stephens and Westoby 2014).

Plants have evolved a myriad of ways to defend themselves against enemies (Koricheva et al. 2004). Investigations into plant defenses and their role in the invasion process have typically focused on resistance traits that directly deter herbivores, such as physical structures (e.g., thorns, spines, trichomes) or secondary metabolites. An additional component of plant defense that is poorly understood in terms of its relation to invasions is tolerance, defined as the maintenance of fitness in damaged plants compared to undamaged plants (Strauss and Agrawal 1999, Boege et al. 2011). Most plants exhibit some capacity to recover from damage, but tolerance is highly variable among species, often depending on plant type (i.e., woody vs. herbaceous), ontogeny, or herbivore identity (Massad 2013). Considering that tolerance allows plants to mitigate the negative consequences of herbivory that they cannot avoid, tolerance might be particularly important for non-native plants in new habitats where they lack specific resistance to novel herbivores. This could be especially important in tropical environments where herbivory rates are relatively high (Coley and Barone 1996).

Investigations into the capacity of invasive plants, either woody or herbaceous, to tolerate damage have been mixed; several studies of single invaders have shown low tolerance (Bossdorf et al. 2004, Huang et al. 2010, Hanley 2012), full tolerance (Rogers and Siemann 2004, Stastny et al. 2005, Callaway et al. 2006, Gard et al. 2013), or overcompensation (Schierenbeck et al. 1994, Lu and Ding 2012). These diverse findings suggest that tolerance may be idiosyncratic among invasive species; however, differing conclusions among single-invader studies may also be attributable to differences in experimental design. Controlled studies of multiple invasive species are less common but are useful because they directly address patterns in species variation and investigate plant traits that may confer tolerance. For example, Ashton and Ler dau (2008) found that the relative growth rate in 12 related temperate vine species following simulated herbivory through leaf clipping was higher in invasive species compared with non-invasive non-natives and native species. After observing high levels of herbivory in the field across all species, including highly invasive species, the authors suggested that high tolerance to herbivory could be a mechanism promoting invader success if plants are unable to avoid herbivory.

Plant tolerance to damage may be driven by a number of underlying traits and induced responses (Tiffin 2000). Induced mechanisms of tolerance function after damage has occurred, often through processes that contribute to regrowth. Examples include activating dormant meristems (Paige and Whitham 1987, Tuomi et al. 1994), reallocating resources from roots to shoots (Hochwender et al. 2012, Rivera-Solís et al. 2012, Barton 2013), increasing photosynthetic rates (Vanderklein and Reich 1999), and increasing growth rates (Nykänen and

Koricheva 2004). On the other hand, constitutive mechanisms of tolerance involve traits that exist prior to damage and aid the plant's ability to manage damage after it has occurred. Assuming growth rate prior to damage is related to growth rate after tissue loss (Weis et al. 2000), plants with high pre-damage growth rates may have the machinery in place for high tolerance. Additionally, constitutive mechanisms of tolerance may depend on biomass allocation before damage such as high root-shoot ratios (Hochwender et al. 2012) and high photosynthetic efficiency (Barton 2016). Tolerance mechanisms may also differ due to ontogeny (Barton 2013). Seedling tolerance is generally predicted to be weak given the limited stored reserves and photosynthetic capacity of seedlings (Strauss and Agrawal 1999) and there is some experimental evidence that damage tolerance can increase significantly even within just a few days during early ontogeny (see Hanley et al. 2004). However, seedling tolerance varies among species, and there is evidence that plasticity in biomass allocation and high rates of photosynthesis can enhance seedling tolerance of herbivory, and that fast growing, weedy species can often tolerate high levels of damage, quickly catching up in size with undamaged control plants (Barton 2008, 2013, Barton and Koricheva 2010). For seedlings, storage tissues in seeds may also contribute to herbivory tolerance, leading to a positive relationship between seed size and tolerance (Bonfil 1998) because larger seeded species generally have larger seedlings, allowing for better access to limiting resources in the environment (plant size effect; Leishman et al. 2000) and larger seeded species may have a relatively large amount of storage reserves, manifested in endosperm, the hypocotyl, or cotyledons during early development (reserve effect; Westoby et al. 1996).

Even though we have learned a great deal about tolerance from previous studies, we still lack unifying concepts that could aid in addressing the role of tolerance in invasion success. Our overall objective was to test the association of herbivory tolerance with plant invasion success using woody plants in Hawaii as a model system. Hawaii has experienced unprecedented rates of plant invasions, with more than 8,000 non-native plant species occurring in both cultivated and natural or semi-natural settings (Staples et al. 2000). Many more species have probably been introduced, but have failed to establish sustainable populations or spread into native communities. Thus, it remains unclear whether certain traits contribute to the successful establishment of some non-native species while limiting the success of others. Herbivory may play a key role in this process. Most native Hawaiian herbivores are extinct or severely reduced in abundance (James 1995). However, Hawaii now has an abundance of introduced herbivores, including mammals, insects, and mollusks, which are consuming plants and pose serious threats to the native flora (Scowcroft and Giffin 1983, Howarth 1985, Joe and Daehler 2008). Because many of these introduced herbivores particularly target the seedling stage (Joe and Daehler 2008), and because seedling

recruitment is a critical aspect of population establishment and spread (Fenner 1987), we focus on herbivory tolerance in seedlings. The goals of our study were to test whether tolerance to herbivory is associated with plant invasion success and to identify plant traits associated with seedling tolerance to herbivory across a range of woody species. We predicted that high relative growth rate, root-shoot ratio, cotyledon mass fraction, and larger seed size will enhance seedling tolerance because of the importance of stored reserves in regrowth following damage. To address these goals, we performed a greenhouse experiment simulating herbivore damage in sixteen non-native woody plant species that differ in invasion status (invasive vs. non-invasive). Because measuring fitness directly in long-lived woody plant species is challenging, we used plant growth and survival, essential components of seedling performance, as metrics of fitness (Barton 2013).

METHODS

Species selection and location

Seedling herbivory is particularly important in Hawaiian mesic and wet forests where non-native herbivorous mollusks are abundant (Stone 1985, Joe and Daehler 2008), and so invasive and non-invasive plant species adapted to tropical mesic or wet conditions were selected for this study (Table 1). We designated species as invasive or non-invasive (Richardson et al. 2000) using local floras (Little et al. 2003, Staples and Herbst 2005), reports of extent of spread (Staples et al. 2000, Imada 2012), and consultations with invasive plant species experts. All of the examined invasive species are widely

naturalized on Oahu and generally across multiple Hawaiian Islands. The selected non-invasive species have been planted around the islands for at least 50 yr (Bishop Museum Herbarium; nsdb.bishopmuseum.org), but do not appear to be spreading or naturalizing. In some cases, the non-invasive species exist only in cultivation. Fruits and seeds of all species were collected from the island of Oahu, primarily from roadside plants and easily accessible trails and landscaping surrounding Manoa Valley. Due to constraints of fruiting seasons, seeds of *Grevillea robusta* and *Cestrum nocturnum* were purchased from nurseries located outside of Hawaii. These two species were not outliers in our tolerance analyses (see *Results*), thus the effect of collecting from a different source population is likely minimal.

The experiment was conducted in a glasshouse on the University of Hawaii Manoa campus from June to December 2013. When possible, species in the same family were planted at the same time, but because time to germination and development of foliar leaves varied among species, experimental treatments were sometimes conducted at slightly different times (up to 20 d apart). Glasshouse mean day and night temperature were 26.8°C (SD = 4.2) and 22.4°C (SD = 1.7), respectively throughout the experiment. Mean seed mass was measured for each species prior to the experiment ($n = 20$ seeds). Because of the small size of *Melaleuca quinquenervia* and *Callistemon viminalis* seeds, an average mass of 50 seeds was used ($n = 5$ replicates). Between 200–1,000 seeds for each species were planted at the start of the experiment in seed trays filled with a 2:1:1 media mixture of Pro-Mix BX (75–85% Canadian sphagnum peat moss, perlite, dolomitic and calcitic limestone, macro- and micronutrients, *Glomus intraradices* mycorrhizae inoculum), 3/4"

TABLE 1. Seedlings of 16 non-native woody plant species from six families and two invasion categories used in a greenhouse experiment to measure tolerance to simulated herbivore damage.

Family	Invasive category	Species	Native range	First recorded in Hawaii
Fabaceae	Invasive	<i>Acacia confusa</i> Merr.	Taiwan and Philippines	1923
		<i>Leucaena leucocephala</i> (Lam.) de Wit	Tropical America	1837
	Non-invasive	<i>Tamarindus indica</i> L.	Tropical Africa and Asia	1919
		<i>Cajanus cajan</i> (L.) Millsp.	Asia	1917
Solanaceae	Invasive	<i>Cestrum nocturnum</i> L.	Central America	1928
	Non-invasive	<i>Solanum melongena</i> L.	India	1824
Bignoniaceae	Invasive	<i>Spathodea campanulata</i> P. Beauv.	West Africa	1915
	Non-invasive	<i>Tabebuia aurea</i> (Manso) Benth. & Hook. f. ex S. Moore	Brazil	1945
Apocynaceae	Invasive	<i>Alstonia macrophylla</i> Wall. ex G. Don	Southeast Asia	1949
		<i>Thevetia peruviana</i> (Pers.) K. Schum.	Central America	1920
	Non-invasive	<i>Nerium oleander</i> L.	Mediterranean, Western Asia	1928
		<i>Carissa macrocarpa</i> (Eckl.) A. DC.	South Africa	1925
Myrtaceae	Invasive	<i>Melaleuca quinquenervia</i> (Cav.) S.F. Blake	Western Australia	1924
	Non-invasive	<i>Callistemon viminalis</i> (Sol. ex Gaertn.) G. Don	Eastern Australia	1933
Proteaceae	Invasive	<i>Grevillea robusta</i> A. Cunn. ex R. Br.	Northern Australia	1880
	Non-invasive	<i>Macadamia integrifolia</i> Maiden & Betche	Eastern Australia	1909

Note: The native range and date first recorded in Hawaii are listed for each species.

granite chips, and perlite respectively. Plants were shaded with 30% aluminum shade cloth and watered four times daily using a misting system, such that the soil mixture did not dry. Germination was recorded when cotyledons appeared. Seedlings were allowed to grow until they developed two foliar leaves, at which time, 45 seedlings per species were transplanted into 10.2 cm × 10.2 cm × 35.6 cm 2.83 L containers (Tall One Treepot, Stuewe & Sons, Tangent, Oregon, USA) filled with the same media mixture as above. Prior to transplanting, 2.5 mL of slow release fertilizer pellets (Osmocote) were added to the surface of each filled pot and mixed with the top 4 cm of the soil. Plants that died within 3 d of transplanting were replaced. Additionally, all pots from different species and treatments were randomly interspersed every 3–4 d on three neighboring glass house benches over the duration of the experiment.

Treatments

Seedlings were randomly assigned to one of four treatment groups (Pre-damage, Control, Defoliated, and Defoliated-JA). The total sample size included 45 plants per species ($n = 11\text{--}12$ per treatment group) × 16 species, for a total of 720 plants. Two groups, Defoliated and Defoliated-JA, had half of each leaf removed by cutting them across the midvein with scissors to simulate herbivory. Cotyledons were left intact. Because mechanical damage alone may not induce the full response to herbivory by animals, Defoliated-JA plants were also sprayed until leaves were dripping with a 0.5 micromol solution of jasmonic acid (JA), a jasmonate derivative. Jasmonates are signaling hormones induced by herbivore attacks and involved in the signal transduction pathways that lead to increases in plant resistance and tolerance following damage (Heil and Ton 2008). Combining mechanical damage with the application of jasmonic acid has been shown to elicit plant responses that are similar to natural herbivory (Constabel et al. 2000, Heil et al. 2001, Zas et al. 2011), which is especially useful when herbivores are unavailable or difficult to work with under controlled conditions (Babst et al. 2005, Hummel et al. 2007). To control for spraying effects, the Defoliated treatment group was sprayed with water. Pre-damage and Control treatments were not mechanically damaged. Pre-damage plants were harvested the same day mechanical damage was applied to Defoliated and Defoliated-JA treatments and were used to obtain pre-damage biomass data. Control plants were also sprayed with water to mimic spraying in the jasmonic acid treatment and were harvested with Defoliated and Defoliated-JA plants 2 weeks later. Upon harvest, aboveground biomass was collected and separated into cotyledon, leaf, and stem portions. Belowground biomass was harvested by cleaning roots of any debris and washing with water. Harvested plant parts were dried in a drying oven at 60°C for a minimum of 1 week or until constant mass, which was measured with a precision of 0.00001 g (Mettler Toledo New Classic MF).

Tolerance indices and pre-damage traits

Since we were interested in evaluating whole plant responses to herbivory, each species' ability to compensate for lost tissue at the end of 2 weeks was determined by calculating a tolerance index (T_i) following Strauss and Agrawal (1999):

$$T_i = M_d/M_c$$

where M_d and M_c are the mean total biomass of plants in the damaged treatments and control treatment respectively. Separate tolerance indices were calculated for the Defoliated and Defoliated-JA treatments for each species.

Several pre-damage biomass traits were measured for each species from the seedlings harvested at the time the damage treatments were applied, including root-shoot ratio, cotyledon mass fraction, and relative growth rate (RGR):

$$\text{RGR} = (M_t - M_s)/T_p$$

where M_t is the mean total biomass of pre-damage seedlings, M_s is the mean seed mass, and T_p is the number of days between planting and the pre-damage treatment biomass harvest. Cotyledon mass fraction was calculated for each species by dividing the cotyledon mass of a seedling by the total biomass of that seedling and averaging among seedlings of that species.

Analysis

To determine if plant invasiveness could explain damage response, we conducted two analyses using linear mixed effects models, following a "step down" approach for significance testing (West et al. 2007). All statistical analyses were performed using SAS Institute Inc 2013, Version 9.4 for Windows (2013). First, we compared above- and belowground growth in invasive and non-invasive species. The dependent variables, above- and belowground biomass, were log transformed to meet normality and homoscedasticity assumptions. The main effects of damage type (Defoliated, Defoliated-JA, and Control), and invasion category (Invasive or Non-invasive) were treated as fixed factors. Tests of the interaction between invasion category and damage type assessed whether invasion is linked with damage tolerance. Plant family and species nested within family were included as random factors. The interaction between species and treatment was also included to determine if compensation patterns were species-specific.

Next, we investigated the ability of plant invasiveness to predict species level tolerance (i.e. tolerance indices) and examined pre-damage traits that could be driving seedling tolerance. Invasion category (Invasive or Non-invasive) was included in the mixed model as a fixed factor. Pre-damage traits (e.g. seed mass, pre-damage root-shoot ratio, pre-damage relative growth

rate, and pre-damage cotyledon mass fraction) were treated as fixed continuous explanatory variables in the model. The effect of plant family was included as a random factor and although not significant, it was retained to control for the effects of phylogeny. Separate mixed model analyses of tolerance indices were conducted for Defoliated and Defoliated-JA treatments.

For mixed model analyses, appropriate diagnostic considerations were made including normal quantile-quantile plots of residuals and scatter plots of predicted values vs. residuals. Outliers were identified, but removing them did not alter conclusions so they were left in. Likelihood ratio tests using a chi squared distribution were used to assess significance of effects based on maximum likelihood for fixed effects and restricted maximum likelihood values for random effects (West et al. 2007). AICc values were also investigated but the results were similar to those of the likelihood tests and are not reported. For all mixed models, the covariance structure type was variance components. Other covariance structures were investigated but did not significantly change the model so are not included in the results. Model construction is summarized in the appendices (above- and belowground growth: Appendix S1; tolerance indices: Appendix S2). Post hoc Tukey adjusted least square mean comparisons examined pairwise differences between significant fixed effects with more than two levels. For analyses of tolerance indices, diagnostics for collinearity among the pre-damage traits were investigated using Variance Inflation Factor (VIF) values from linear regression analyses. Pre-damage seed mass and relative growth rate had high VIF values ($VIF > 10$; Appendix S3: Table S1), suggesting multicollinearity. To avoid redundancy, seed mass was not included in mixed model analyses; however, relative growth rate was retained to provide information about physiological traits at the time of defoliation that may confer tolerance. Removing seed mass reduced VIF values considerably for remaining variables ($VIF < 3.3$; Appendix S3: Table S1).

Because we were also interested in mechanisms that may drive overcompensation following herbivory, for species

with a tolerance index greater than one, we investigated relative biomass allocation in terms of root-shoot ratio between Defoliated and Defoliated-JA treatments using a *t*-test. Appropriate diagnostic considerations were made to ensure normality and equality of variance.

RESULTS

Invasion category and tolerance

Contrary to our prediction, invasive species did not tolerate seedling damage better than non-invasive non-natives, as evidenced by a non-significant invasion category $\times$ treatment interaction for growth (Table 2), and tolerance indices that did not significantly differ between invasion categories for either Defoliated or Defoliated-JA treatments (Table 3). There was also no significant effect of invasion category on mean plant biomass across treatments (Table 2; Fig. 1). Although the effect of invasion category was not significant, defoliation treatments did affect both above- and belowground biomass (Table 2). Aboveground biomass of Control plants was significantly greater than that of Defoliated (post-hoc Tukey $t = -4.54$, $P < 0.001$) and Defoliated-JA (post-hoc Tukey $t = -7.25$, $P < 0.001$) treatment plants. Plants in the Defoliated treatment had significantly more aboveground biomass than Defoliated-JA plants (post-hoc Tukey $t = -2.69$, $P = 0.020$). Similar patterns were detected for belowground biomass between damage and controls, with significantly more belowground biomass in Control plants than plants from either Defoliated (post-hoc Tukey $t = -3.00$, $P = 0.010$) or Defoliated-JA (post-hoc Tukey $t = -5.07$, $P < 0.001$) treatments. However, in this case, there was no significant difference between the two damage treatments (post-hoc Tukey $t = -2.07$, $P = 0.105$). Species nested within family was significant for both above- and belowground biomass (Table 2). All but two damaged seedlings (*C. viminalis* Defoliated group) of all species survived. Among Control plants, a single *M. quinquenervia* seedling did not survive.

TABLE 2. Effects of invasion category (invasive vs. non-invasive) and defoliation treatment (control vs. 50% defoliation vs. 50% defoliation + jasmonic acid application) on log transformed above- and belowground biomass measured 2 weeks after defoliation.

	Log aboveground		Log belowground	
	Test statistic $\chi^2 (I)$	P-Value	Test statistic $\chi^2 (I)$	P-value
Fixed effects				
Invasion category	$\chi^2 (1) = 1.1$	0.294	$\chi^2 (1) = 1.3$	0.254
Treatment	$\chi^2 (1) = 28.9$	<0.001	$\chi^2 (1) = 19.3$	<0.001
Invasion category $\times$ Treatment	$\chi^2 (1) = 3.3$	0.069	$\chi^2 (1) = 3.0$	0.083
Random effects				
Family	$\chi^2 (2:1) = 0.8$	0.521	$\chi^2 (2:1) = 3.4$	0.123
Species (Family)	$\chi^2 (2:1) = 1023.4$	<0.001	$\chi^2 (2:1) = 739.1$	<0.001
Species $\times$ Treatment	$\chi^2 (3:2) = 2.6$	0.969	$\chi^2 (3:2) = 1.9$	0.951

Notes: Species ($n = 16$) nested within plant family and family are treated as random effects. Linear mixed models results are shown. Bolded variables are significant at $\alpha = 0.05$.

TABLE 3. Effects of pre-damage traits and invasion category on tolerance of seedlings of sixteen non-native woody species 2 weeks after simulated herbivory with and without jasmonic acid application.

	Defoliated			Defoliated-JA		
	Estimate (SE)	Test statistic	P-value	Estimate (SE)	Test statistic	P-value
Fixed effects						
Cotyledon mass fraction	0.755 (0.380)	$\chi^2 (1) = 5.0$	0.025	1.157 (0.360)	$\chi^2 (1) = 9.4$	0.002
Relative growth rate	0.262 (0.751)	$\chi^2 (1) < 0.1$	1.0	1.007 (0.719)	$\chi^2 (1) = 2.6$	0.107
Root-shoot ratio	0.702 (0.370)	$\chi^2 (1) = 4.7$	0.030	0.762 (0.342)	$\chi^2 (1) = 4.7$	0.030
Invasion category	–	$\chi^2 (1) < 0.1$	0.655	–	$\chi^2 (1) = 0.1$	0.752
Invasive	0.417 (0.178)	–	–	0.220 (0.169)	–	–
Non-invasive	0.416 (0.201)	–	–	0.247 (0.191)	–	–
Random effects						
Family	0.011 (0.013)	$\chi^2 (0:1) = 2.0$	0.079	0.017 (0.016)	$\chi^2 (0:1) = 1.8$	0.090

Notes: Species level tolerance was calculated using a tolerance index (mean biomass of damaged plants divided by mean biomass of controls) for Defoliated and Defoliated-JA treatments. Linear mixed model results are shown. Estimates of parameter coefficients for fixed and random effects are included. Bolded variables are significant at $\alpha = 0.05$.

Tolerance indices and pre-damage traits

Most of the species in the experiment failed to tolerate the damage completely as evidenced by tolerance indices < 1.0 (Fig. 2). However, *Alstonia macrophylla*, *Thevetia peruviana*, *Solanum melongena*, and *Macadamia integrifolia* nearly tolerated the Defoliated treatments (Fig. 2). In both Defoliated and Defoliated-JA treatments, *C. viminalis* plants overcompensated in terms of total biomass (Fig. 2). Mean root-shoot ratio in *C. viminalis* was greater in the Defoliated-JA treatment than in the Defoliated

treatment (mean $\pm$ SE = 0.93 ± 0.07 and 0.62 ± 0.07 , respectively; $t = -3.02$, df = 18, $P = 0.004$).

Cotyledon mass fraction and root-shoot ratio had significant positive effects on tolerance in both Defoliated and Defoliated-JA treatments, revealing these traits as important mechanisms enhancing tolerance (Table 3). In contrast, relative growth rate did not explain a significant amount of variance in tolerance and was not included in the final model for either Defoliated or Defoliated-JA treatments (Table 3). The random effect of plant family was also not significant but was retained in the final model to control for phylogeny (Table 3).

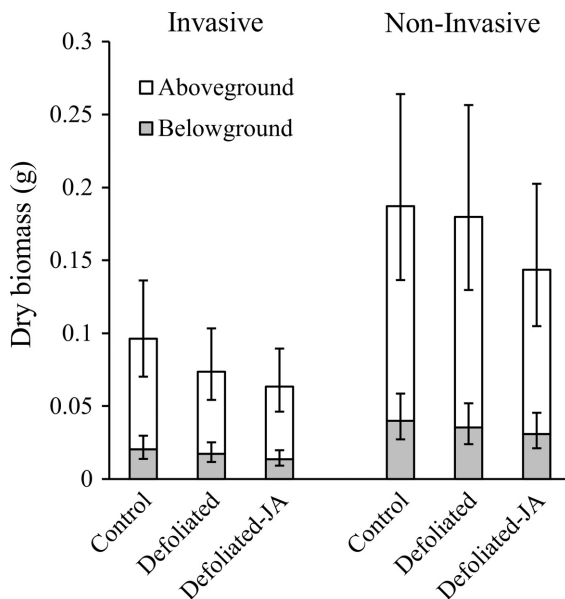


FIG. 1. Above- and belowground biomass across seedlings of 16 non-native woody plant species from six families and two invasion categories (invasive and non-invasive) 2 weeks after 50% foliar removal with or without jasmonic acid application (Defoliated and Defoliated-JA). Control plants received no defoliation. Biomass is least square mean. Bars are 95% Confidence Intervals.

DISCUSSION

Invasions and tolerance

In our first investigation of the role that tolerance to seedling herbivory might play in woody plant invasions, we detected no evidence that tolerance to simulated herbivory is higher in invasive than non-invasive species. In general, most species examined were unable to fully tolerate herbivory within the two-week period allowed, which may indicate that seedling herbivory could be a constraint to most of these species in Hawaii. However, five species nearly fully tolerated damage, and a single species showed overcompensation, highlighting species variation. In the only other multiple invader study comparing invasive and non-invasive species, Ashton and Lerdau (2008) found evidence that tolerance may play a role in the success of temperate woody vines in North America, where invasive species tolerated damage more than non-invasive or native vines. Even though our experimental design allowed us to isolate and assess tolerance as a potential invasion barrier, there is a need to revisit this type of study in the field to determine how well these results mirror responses to herbivory under natural conditions as opposed to ideal greenhouse conditions (Hanley and Fegan 2007).

The majority of the studies that have experimentally tested tolerance to herbivory in non-native species

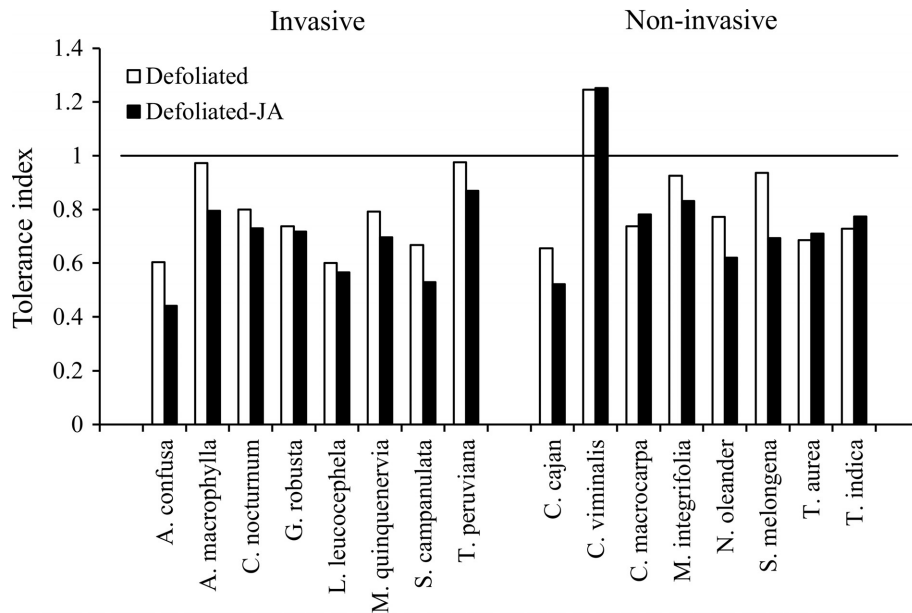


FIG. 2. Species level seedling tolerance indices (mean biomass of damaged plants divided by mean biomass of controls) of invasive ($n = 8$) and non-invasive ($n = 8$) woody species 2 weeks after 50% foliar removal with or without jasmonic acid application (Defoliated and Defoliated-JA). Tolerance indices >1 represent undercompensation, and indices <1 represent overcompensation.

measured plant performance after a much longer recovery time than in our study (1–6 months). Thus, it is possible that invasive species in our study did not have enough time to fully compensate for damage. For example, Hanley (2012) observed that *P. lanceolata*, a non-native invasive perennial herb in North America, did not fully compensate for damage in terms of biomass after 21 d, but that reproductive traits in damaged plants, including time to flower and the number of inflorescences, were similar to controls after 150 d. This study highlights that the metric of fitness used to quantify tolerance may affect interpretations (see also: Hanley and May 2006, Hanley and Fegan 2007). In our study on woody plants, it was not feasible to measure tolerance to seedling damage with respect to reproductive output due to long generation times. However, given the importance of fast growth during the seedling stage for establishment among competitors in nature, growth rate and biomass are important early metrics of fitness. A delayed tolerance response has been noted in other studies (e.g. Haukioja and Koricheva 2000, Sacchi and Connor 2013), indicating that short-term studies may in general underestimate tolerance responses.

Given that the majority of highly invasive species in our study failed to fully compensate for damage and that low tolerance has been found in other invasive species (e.g., Bosssdorf et al. 2004, Huang et al. 2010, Hanley 2012), understanding why these species may still be invasive is important. First, plants may possess chemical or structural defense traits (i.e., resistance or avoidance traits) that reduce or prevent herbivory. For example, *C. nocturnum* (invasive: Solanaceae), produces steroidal

saponins and tannins in leaf tissue (Mimaki et al. 2001, Patil et al. 2011) and *G. robusta* (invasive: Proteaceae) produces hydrogen cyanide in flowers and fruits (Ullah et al. 2014). Saponins, tannins, and cyanide are known anti-herbivore secondary metabolites (Wink 2003) that may make these species less palatable. Thus, having low tolerance may not affect invasion success if these plants are experiencing low levels of herbivory in the field. Second, plants may “escape” herbivory by recruiting to safe sites. Subsequent high growth rates of seedlings in safe sites, or high reproductive capacity of plants that recruit to safe sites could make up for fitness losses due to low tolerance in seedlings outside of safe sites. This is certainly possible with *G. robusta* (invasive: Proteaceae), *Spathodea campanulata* (invasive: Bignoniaceae), and *M. quinquenervia* (invasive: Myrtaceae). These species are prolific in terms of wind-dispersed seeds and have high germination rates in the lab (unpublished data), which may increase chances of finding, and recruiting to, safe sites. Similarly, high propagule pressure has been suggested as a mechanism to overcome invasion barriers, especially in association with establishment (Simberloff 2009). For example, less tolerant species may be able to overcome biotic resistance through repeated introductions that make seedlings more likely to find safe sites. This seems possible for *M. quinquenervia* (invasive: Myrtaceae), which is commonly used as a landscape tree in urban Oahu.

Application of jasmonic acid reduced the growth of aboveground biomass compared to mechanical wounding alone. Combining mechanical damage with the application of jasmonates, including the derivative jasmonic acid,

has been shown to be an effective way to elicit plant responses that are similar to natural herbivory (Constabel et al. 2000, Heil et al. 2001, Zas et al. 2011). One possible explanation for our results is that jasmonic acid induced changes in resistance traits (such as secondary compounds and physical defense traits) at the expense of compensatory growth. Additional studies incorporating bioassays with appropriate herbivores and measurements of resistance traits would reveal whether simultaneous induction of tolerance and mechanisms aimed at reducing herbivore damage is occurring in these species.

Overcompensation in C. viminalis

The only species in our study to overcompensate for damage was the non-invasive *C. viminalis*. Overcompensation in terms of total biomass occurred in both damage treatments (Defoliated and Defoliated-JA). Interestingly, plants with jasmonic acid application allocated more biomass to roots, evidenced by a higher root-shoot ratio, than plants without jasmonic acid. Assuming jasmonic acid application induces natural responses to herbivory, then it could be that herbivory induces sequestration of resources into belowground tissues, which may be used some time following herbivory for additional aboveground growth (Schwachtje and Baldwin 2008, Oriens et al. 2011). However, previous work on seedlings has generally indicated that growth rates of shoots are enhanced at the expense of roots (Barton 2008, 2013, 2016) likely due to prioritization of carbohydrate mobilization to growing apical meristems (Yoshizuka and Roach 2011, Lucas et al. 2013, Schmidt et al. 2015). Given that this species has not been able to successfully spread well beyond planting sites in Hawaii, despite being able to tolerate herbivory at early growth stages, there are likely other barriers to its invasion success. There is evidence that for several other woody plant species in Hawaii, biotic resistance through competition and herbivory (Bufford et al. 2016), and sterility or lack of pollinator services (Bufford and Daehler 2014), cause non-native species to fail.

Tolerance related to pre-damage allocation of resources

We found that investment of resources in storage organs, specifically roots and cotyledons, prior to damage increased compensation to defoliation across species. This result adds to a growing consensus that pre-damage growth patterns determine tolerance to damage, perhaps even more than induced responses (Hochwender et al. 2000, Myers and Kitajima 2007, Stevens et al. 2008, Rivera-Solis et al. 2012, Barton 2013, 2016). However, very little of this research has explicitly focused on seedlings (but see Myers and Kitajima 2007, Barton 2008, 2013, 2016). In general, seedlings are predicted to have low tolerance to damage because even relatively small herbivores can remove a large portion of biomass relative to the size of the plants they attack (Barton and

Hanley 2013). Yet there is evidence that some species, particularly weedy herbs with fast growth rates, can tolerate herbivory to the same degree as older ontogenetic stages (Barton and Koricheva 2010, Barton 2013). During germination and early development, plants rely heavily on storage reserves such as the seed and cotyledon, so it is not surprising that these types of organs provide resources so that plants can grow following herbivory (Erb et al. 2009). For example, Barton (2013) found that in the herbaceous perennial *Plantago major* L., tolerance to 50% defoliation was related to high root-shoot ratios prior to damage, concluding that root biomass was an important driver of tolerance. Similarly, in a common garden experiment of seven understory species, Myers and Kitajima (2007) found that survival after a year following 100% defoliation of 3–18 week old seedlings was related to the initial amount of non-structural carbohydrates (e.g., starch and simple sugars) within the stems and roots. As in these studies, we found a positive association of tolerance with root-shoot ratio across many different species. Furthermore, we found that species with relatively large cotyledons for their size had higher tolerance to damage, providing additional support for the conclusion that pre-damage investment in storage organs (roots and cotyledons) are key mechanisms of seedling tolerance. Given that cotyledon mass fraction and root-shoot ratio were moderately correlated in our analysis, conclusions about the predictive power of each of these traits is limited due to collinearity. It may be that high cotyledon mass is important for tolerance, but requires a simultaneous investment in roots or vice versa, or that each of the traits is important singly.

Pre-damage relative growth rate (RGR) was not indicative of plant tolerance, contrary to our predictions and previous work (e.g., Myers and Kitajima 2007). Constitutive pre-damage RGR should only be important in tolerance if plants can maintain high levels of growth following damage (Weis et al. 2000) and to assess this, post-damage measurements of RGR are also needed.

CONCLUSION

A suite of plant traits and environmental circumstances is likely to explain why some plants are invasive and others are not (Theoharides and Dukes 2007). Trying to identify unifying patterns is a necessary goal for invasion ecology and is especially important in oceanic islands where plant invasions are a major threat to ecosystems. The findings presented in this paper suggest that greater defensive investment in tolerance to herbivory is not an important driver of invasion success in seedlings of woody plants in Hawaii. Despite considerable variation among species in the extent to which damaged plants re-grew, there were no significant differences in tolerance between invasive and non-invasive species in our study. Given that we only allowed plants to grow for 2 weeks following damage, it may be that invasive species needed longer to fully compensate. However, we

observed overcompensation in one non-invasive species, suggesting that plants can compensate for biomass loss, even over short time periods. Additionally, seedling tolerance to simulated herbivory was enhanced by an increased investment in potential storage reserves, specifically cotyledons and roots, adding to our understanding of defense mechanisms, especially in young plants. The lack of higher tolerance in highly invasive species, relative to non-invasive species, may suggest that invaders may use other traits to overcome herbivory invasion barriers, such as resistance traits (chemical or structural defenses) or escape from herbivores through occupation of safe sites, or that seedling herbivory does not play an important role in the plant invasion dynamics of these woody species in Hawaii.

ACKNOWLEDGMENTS

We thank Danielle Frohlich for her expertise and assistance with non-native species selection. We also thank Jennifer Bufford and Andy Taylor for providing statistical insight. Kari Bogner and Risha Mishima provided invaluable help in the greenhouse and laboratory. Greenhouse space was provided by Dr. Travis Idol, and is managed by the Department of Natural Resources and Environmental Management at the University of Hawaii at Manoa. Finally, we appreciate the insightful comments and suggestions from anonymous reviewers, which greatly improved the quality of our manuscript. This research was made possible by a grant from the University of Hawaii Manoa Graduate Student Organization.

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